

Energy: $|n_x, n_y\rangle$
 $|0,1\rangle, |1,1\rangle$
 $\rightarrow$ Spin: $|l, m_l\rangle \rightarrow$

$$|4\rangle = \frac{1}{2}|2,0\rangle - \frac{1}{2}|0,2\rangle + i\frac{1}{\sqrt{2}}|1,1\rangle$$

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

• $|\psi\rangle = \sum_{n=0}^{\infty} c_n |V_n\rangle$ $\langle x|V_n\rangle = v_n(x)$

$$\psi(x) = \langle x|\psi\rangle = \sum_{n=0}^{\infty} c_n \langle x|V_n\rangle$$

$$\psi(x) = \sum_{n=0}^{\infty} c_n v_n(x)$$

$$\hat{L}_z = -i\frac{\partial}{\partial \phi}$$

$$\hat{L}_z|\psi\rangle = \lambda|\psi\rangle$$

$$\langle x|\psi\rangle = \psi(x) = \frac{1}{2}\langle x|2,0\rangle - \frac{1}{2}\langle x|0,2\rangle + \frac{i}{\sqrt{2}}\langle x|1,1\rangle$$

$\downarrow$
 $v_{2,0}$

$\downarrow$
 $v_{0,2}$

$\downarrow$
 $v_{1,1}$

$$(\hat{L}_z\psi)(x) = -i\frac{\partial}{\partial \phi}(\quad)$$

Term-by-term:

$$\psi(x) = \frac{1}{2} \left[\frac{1}{\sqrt{8\pi}} (4x^2 - 2) e^{-x^2-y^2/2} \right] - \frac{1}{2} v_{0,2} + \frac{i}{\sqrt{2}} v_{1,1}$$

Polar: $x = r \cos(\phi) \quad | \quad x^2 + y^2 = r^2$
 $y = r \sin(\phi)$

$$\underline{1^{st} \text{ term}} : \frac{1}{\sqrt{8\pi}} (2r^2 \cos^2(\phi) - 1) e^{-r^2/2} \quad \left(\frac{1}{2} Y_{2,0} \right)$$

$$\underline{2^{nd} \text{ term}} : - \frac{1}{\sqrt{8\pi}} (2r^2 \sin^2(\phi) - 1) e^{-r^2/2} \quad \left(\frac{1}{2} Y_{0,2} \right)$$

$$\underline{3^{rd} \text{ term}} : \frac{i}{\sqrt{2}} \frac{1}{\sqrt{4\pi}} (4r^2 \cos(\phi) \sin(\phi)) e^{-r^2/2} \quad \left(\frac{i}{\sqrt{2}} Y_{1,1} \right)$$

$$|\psi\rangle = \nearrow$$

$$L_z |\psi\rangle = \lambda |\psi\rangle$$

add all 3:

$$|\psi\rangle = \frac{1}{\sqrt{8\pi}} 2r^2 (\cos^2(\phi) - \sin^2(\phi)) e^{-r^2/2} + \frac{i}{\sqrt{8\pi}} (4r^2 \cos(\phi) \sin(\phi)) e^{-r^2/2}$$

factor:

$$|\psi\rangle = \frac{2}{\sqrt{8\pi}} e^{-r^2/2} \left[(\cos^2(\phi) - \sin^2(\phi)) + 2i \cos(\phi) \sin(\phi) \right]$$

$$\sin^2(\phi) + \cos^2(\phi) = 1$$

$$\underline{\sin^2(\phi)} = 1 - \cos^2(\phi) \quad \nearrow$$

$$\underline{} = \cos^2(\phi) - [1 - \cos^2(\phi)] = 2\cos^2(\phi) - 1$$

$$\cos(2\phi) = 2\cos^2(\phi) - 1$$

$$\cos^2(\phi) - \sin^2(\phi) = \cos(2\phi)$$

$$\underline{} = 2i \cos(\phi) \sin(\phi)$$

$$\sin(2\phi) = 2 \sin(\phi) \cos(\phi)$$

$$= 2i \sin(2\phi)$$

$$|\psi\rangle = \frac{2}{\sqrt{8\pi}} e^{-r^2/2} (\cos(2\phi) + i \sin(2\phi))$$

$$|\psi\rangle = \frac{2}{\sqrt{8\pi}} e^{-r^2/2} e^{2i\phi} \quad * *$$

$$\hat{L}_z |\psi\rangle = -i \frac{\partial}{\partial \phi} \left[\frac{2}{\sqrt{8\pi}} e^{-r^2/2} e^{2i\phi} \right] = 2 |\psi\rangle$$

$$\hat{L}_z |\psi\rangle = \underline{m_l} |\psi\rangle$$

$$\hat{L}^2 |\psi\rangle = l(l+1) |\psi\rangle$$

$$\hat{L}^2 = L_x^2 + L_y^2 + L_z^2$$

(b)

$$\hat{H} |\psi\rangle = E |\psi\rangle \quad : \quad \underline{E = 3}$$

(c) $E = V_{\text{eff}}(r)$

$$\hat{H} = \hat{T} + \hat{V} = \underbrace{-\frac{1}{2r} \frac{d}{dr} \left(r \frac{d}{dr} \right)}_{\text{Kinetic}} + \underbrace{\frac{\overset{M=2}{2^2}}{2r^2} + \frac{r^2}{2}}_{\text{Potential}}$$

$$\frac{2}{r^2} + \frac{r^2}{2} = 3 = E \quad (\text{Solve for } r)$$

(d) $|\psi\rangle = \sum_{n=0}^{\infty} \underline{c_n} |\psi_n\rangle$

$$\langle \psi | \psi \rangle = 1 = \sum_{n=0}^{\infty} |c_n|^2 \quad *$$

Electron: $|c_1|^2 = |c_2|^2 = 1/2$

$$\langle \psi | \psi \rangle = \int_{-\infty}^{\infty} \underline{dx dy} \, \psi^*(x,y) \underline{\psi(x,y)} = 1$$

$$= \int_0^\infty r dr \int_0^{2\pi} d\phi \left(\frac{2}{\sqrt{8\pi}} e^{-r^2/2} e^{-2i\phi} \right) \left(\frac{2}{\sqrt{8\pi}} e^{-r^2/2} e^{2i\phi} \right)$$

$$= \left(\frac{4}{8\pi} \right) \int_0^\infty r dr \underbrace{\int_0^{2\pi} d\phi}_{2\pi} e^{-r^2}$$

$$= \int_0^\infty r e^{-r^2} dr = 1$$

$$= \int_0^a r e^{-r^2} dr + \int_a^\infty r e^{-r^2} dr = 1$$

From (c), there's some r for which is classically forbidden. Suppose $a < r < b$ is this region.

$$\underbrace{\int_0^a r e^{-r^2} dr} + \underbrace{\int_a^b r e^{-r^2} dr} + \underbrace{\int_b^\infty r e^{-r^2} dr} = 1$$

Prob. of being in that region.
(See Page 6 of Notes).

